

Javier Rangel Murillo

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PROFESSIONAL SUMMARY

Backend Engineer with startup experience designing and owning production systems end-to-end — from distributed architecture and cloud infrastructure to payment pipelines and async workers. Specialized in **TypeScript/Node.js** and **Python**, with proven delivery on **GCP**, **Docker**, and event-driven systems. Sole backend author of a live legal-tech platform: 66 API routes, 11 background workers, full payments, identity verification, and video workflows.

TECHNICAL SKILLS

Backend (Node.js, Express, APIs) ■■■■■■

Programming (TypeScript, Python, SQL) ■■■■■■

Cloud & Infra (GCP, Firebase, Terraform) ■■■■□

Databases (PostgreSQL, Redis, MongoDB) ■■■■□

Async Systems (BullMQ, Pub/Sub, Workers) ■■■■□

DevOps (Docker, CI/CD, GitHub Actions) ■■■■□

Integrations (PayPal, Twilio, SendGrid) ■■■■□

Developer Tools (Linux, Git, Puppeteer) ■■■■■■

PROFESSIONAL EXPERIENCE

LegalTec

Founding Backend Systems Engineer | Oct 2025 – Mar 2026

- Architected and shipped the entire backend platform using **Node.js**, **TypeScript**, and **Express.js** as **sole author of 91.6%** of the codebase — 66 API routes, 78+ services, 11 background workers — taking a legal-tech startup from prototype to production.
- Designed **GCP infrastructure** (Cloud Run, Pub/Sub, Tasks, KMS, Secret Manager, Firebase Auth/Firestore) with encryption at rest, enabling high-availability distributed processing of sensitive legal data.
- Built 11 specialized workers with **BullMQ** + **Redis** (payment retry, hold cleanup, video ETL, recording TTL, Google Meet processing, verification pipelines), eliminating synchronous bottlenecks across all async workflows.
- Delivered full platform monetization via **PayPal APIs** (Orders, Vault, Subscriptions, Webhooks) with idempotent retry logic; automated virtual appointments via **Google Workspace/Meet API** (event lifecycle, video ETL).
- Engineered **Puppeteer**-based scrapers for CURP/RENAPO and SEP Cédula Profesional validation, replacing third-party API dependencies and cutting identity verification costs to zero.
- Delivered production observability: **Prometheus** metrics, **Pino** structured logging, audit trail, circuit breakers (Oposum), rate limiting, and a 25-file **OpenAPI** documentation suite.
- Provisioned **Terraform** IaC on GCP; configured **GitHub Actions** CI/CD pipelines to Cloud Run for staging and production with zero-downtime deploys.

Universidad Panamericana – iOS Development Lab

Backend Lead | Oct 2022 – Present

- Promoted to lead Sep 2025; mentors students on **Python**, clean architecture, and API design while guiding projects from monoliths to containerized microservices with **Docker**.
- Built and deployed production APIs with **Django** and **Flask** backing native Swift iOS apps; designed **PostgreSQL** and MongoDB schemas for performance and data security.

PROJECTS & ACHIEVEMENTS

Notadio (Express, TypeScript, Python, whisper.cpp, Ollama, FFmpeg)

- Built a **local-first transcription service** — processes audio/video with **whisper.cpp** (zero API cost) and generates AI summaries via **Ollama** using a MapReduce chunking pipeline. Speaker diarization (**pyannote**), multi-format export, live telemetry; deployed on VPS with Caddy + systemd.

backwell-api (Rust, Actix-web, Django, PostgreSQL, Docker)

- Dual-language microservices system for **combinatorial schedule optimization**. **Rust/Actix-web** handles high-performance graph algorithms (petgraph) for conflict-free course scheduling; **Django/DRF** manages domain CRUD. Containerized with Docker Compose.

Redes que Cambian (Django, Angular, PostgreSQL, Docker)

- Full-stack platform connecting individuals with NGOs across Mexico — volunteer coordination, project management, social-impact metrics — deployed for a live non-profit organization.

EDUCATION

Universidad Panamericana Mexico City, Mexico

B.Eng. in Data Intelligence and Cybersecurity

Aug 2022 – Jun 2027

Languages: Spanish (native) • English C1 – TOEFL iBT 607